Theft through USB drives
Companies should encrypt data on their hard drives as well, so employees cannot walk out with sensitive documents on their USB drives

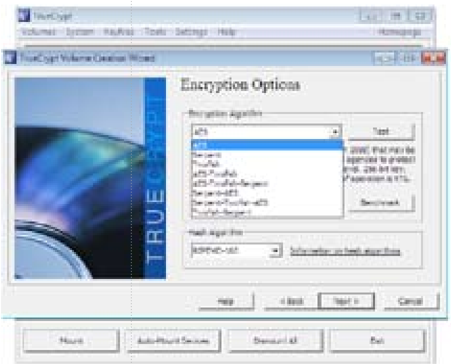
Corporate Theft
Studies show that 45% of employees steal data while changing jobs, are often hired by competitors

Selecting a location for encryption

a drive in TrueCrypt, the computer will recognise a plugged in drive, but not allow you to access it. Instead, any computer will prompt you to format the drive before you can use it. TrueCrypt can be used to mount the drive, which means that the same drive will occupy to drive letters. The default one will be inaccessible and the mounted drive will be available only after you enter the password in TrueCrypt. TrueCrypt is available for Linux and Mac systems as well, so your encrypted drive will be truly portable.

To encrypt a drive, plug it into the computer, start up TrueCrypt, and go to `Volumes>Create New Volume`. A volume is a location where encrypted and password protected data can be stored securely. This starts up the TrueCrypt Volume Creation Wizard. There are three possible locations for the data to be stored. The first option is to create an encrypted file container. This option allows you to create an image file, which stores all the data encrypted; you can write any kind of data inside this image file. This image file can be carried around in portable devices. The second option lets you secure an entire portable device, including USB drives and portable hard drives. This option is Encrypt a non-system partition / drive. The third option is to encrypt an entire partition or a system drive. Choose the second option and click on next.

The wizard then offers you two options for the type of volume. The first option is a Standard TrueCrypt volume, the second option is a Hidden TrueCrypt volume. The Hidden TrueCrypt volume adds another layer of security. If for any reason, anyone finds out



Selecting a cipher, or a combination of ciphers

that you have encrypted your portable drive, and force you to reveal it using a password, then the Hidden TrueCrypt volume will come to your rescue. You can hide sensitive looking data in the TrueCrypt volume, but the real sensitive data will be stored in the hidden volume,

CIPHERS IN TRUECRYPT

All the options available in TrueCrypt are block ciphers. Block ciphers take pieces of the input data, and convert it into pieces of coded data. The pieces range from 16 bits to 256 bits. The different encryption options available are different methods of substitution. The AES is the most commonly used encryption algorithm, and is the standard for the official purposes of many governments. The other options available include Twofish and Serpent. Serpent is more secure than AES, but is understandably slower. Users can use any combination of the three ciphers, or use all three together (very slow).

which leaves absolutely no traces as it is located invisibly within the encrypted volume. This is for advanced users, and not really necessary in most cases. Simply select the Standard TrueCrypt volume and proceed. The next step is to select the drive or device that you want to encrypt and password protect. Click on Device, and select a device from the list that appears, then click on Next.

The next step allows you to either format the device and encrypt it, or encrypt the partition in place. The first option simply erases all the data on the partition, then encrypts it. The second option, encrypts the drive, moves the data around, and encrypts the data as well. The second option is very time consuming, and it is faster to copy the data to a secure location, format and encrypt the drive, then copy the data back to the volume. Select Create encrypted volume and format it, then click on Next.

Mounting the encrypted volume

This is where things get a little geeky. You get to choose from a range of encryption ciphers. There are a range of options available. Generally speaking, the more the time displayed, the more secure the ciphers are. See box for more details. Once an encryption method for a drive has been selected, it cannot be changed later on. Click on Next.

The size prompt box now appears. This is useful only for creating containers, or volumes hidden within a file. The size of the drive will show up in this window, and you cannot change the values. Click on Next. You have to enter a password and confirm it at this stage. If you use anything less than a password of 20 strings in length, TrueCrypt will prompt you to choose a more secure passwords.

A ten character password or so should be beyond what a casual snooper can crack using a brute force method, which is trying every possible combination of letters and numbers as the password. Next you will be asked to move the cursor around the window in a random way. This just generates random strings of data that will be used in the encryption. Click on Format after this is done.

If there is data on your device, you will be prompted about the loss of the data. Note that the drive will be formatted, and all the data overwritten using garbage values. Even a file recovery software cannot recover your data after you proceed. Click on Next to continue. TrueCrypt will now erase all the data, and encrypt the external drive. Even with the default formatting, and the fastest encryption algorithm, the process will take about five minutes for each GB. It is a good idea to leave the process overnight, if you are encrypting a portable hard drive. The portable version of TrueCrypt can mount volumes, but not create them. To mount a volume, select a drive letter from the list of available drive letters. Then click on Mount, and select the drive as it shows up in the system's drive list. You will be prompted for a password, which you have to enter at this point of time. Now the drive will show up in explorer. You can use it as any other drive, copy and paste data from it, but without TrueCrypt, and your password, it cannot be opened by anyone else.